

A Study of the Mixing Index in Solid Particles[†]

Shih-Hao Chou, Yue-Lou Song and Shu-San Hsiau*

[†] Department of Mechanical Engineering, National Central University, Taiwan

Abstract

This study focuses on investigating the mixing state by different mixing indices for different pre-organized particles. We also try to propose a new mixing index and discuss the advantages and disadvantages of different cases.

The results show that the Avg. distance mixing index indicates the degree of mixing more accurately than other mixing indices for a case of complete segregation. In some cases the degree of mixing would be higher with the Lacey mixing index compared with the other indices, with the size of the cells greatly affecting this value. In other words, the use of any single specific mixing index on its own cannot accurately reflect the correct degree of mixing for all different kinds of systems. The only difference between mixing indices is which one can provide more accurate and stable results.

Keywords: mixing index, degree of mixing, granular material, pre-organized condition

1. Background

The mixing of powders is very important in industry as well as in our daily life. Many products or semi-products are made from different ingredients in the form of powders which need to be well mixed to make ensure a high quality end product. For example, for manufacturing pharmaceuticals, avoiding segregation and achieving a well-mixed powder is essential. Thus, the problem of mixing/segregation in powdered or granular materials has been studied by many industrial engineers and academics.

Powders can be mixed by mechanical excitation, gas-flow agitation, gravity-driven free-falling, and many other different complex methods (Masuda H. et al., 2006). Different types of blenders/mixers have been developed for industry. The mechanisms of powder mixing include convective mixing, shear mixing and diffusive mixing (Lacey P.M.C., 1954; Kaye B.H., 1997; Ottino J.M. and Khakhar D.V., 2000). The rate of mixing by diffusive mixing is low compared with convective mixing mechanisms. However, diffusive mixing is essential for microscopic homogenization (Gotoh K. et al., 1997).

What is a so-called “well-mixed” granular material? The mixing status is usually determined based upon one or several samples taken from a batch. The way the sam-

ple is taken, the number of samples, the locations the samples are taken from, the size of the samples, and so on, all influence the evaluation of the mixing status (Weidenbaum S.S., 1958). The statistical method for investigating the samples is also critical. This paper studies determination of the mixing index for particle mixing by sampling.

As mentioned above, the mixing index is an important parameter to indicative of the status of mixing, with a value of 1 indicating that the material is fully mixed; 0 represents fully segregated. However there are many different mixing indices with different ways of determining the status some of which are investigated in this study to determine which one is the best and the most user friendly?

2. Introduction to some mixing indices

Currently, there are a number of suitable mixing indices that can be used to analyze granular mixing behavior. Gayle J.B. et al. (1958) proposed a mathematical method to solve the mixing problem based on the chi square value. Their method can not only be used for a binary mixture system but also to for systems with any finite number of components. Leggatt C.W. (1951) studied the mixing process for weeds and crops, and determined the dispersion of homogeneity in a unit weight. Poux M. et al. (1991) also made a careful review of many different mixing indices.

Most investigations of mixing indices are based on the

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[†] No. 300, Zhongda Rd. Zhongli City, Taoyuan County 32001, Taiwan

* Corresponding author: Shu-San Hsiau;

E-mail: sshsiau@cc.ncu.edu.tw

TEL: +886-3-426-7341 FAX: +886-3-425-4501

statistical method but photometric and kinetic approaches are also employed. However, the conditions for power mixing are defined differently in different mixing indices indicating the uncertainty and complicated nature of the mixing process. In this study, some of the mixing indices commonly used in academic research or in industry are reviewed; only binary mixtures are considered.

(1) Lacey Mixing Index

Lacey P.M.C. (1943) first introduced what is now known as the Lacey mixing index in 1943, by dividing the area of interest into N cells. Assume that ϕ_i is the concentration of the reference component in the “ i th” cell and ϕ_m is the overall concentration of the reference component. The variance of σ^2 for the concentration of a reference component in each cell can be expressed by

$$\sigma^2 = \sum_{i=1}^N \frac{(\phi_i - \phi_m)^2}{N-1}, \quad (1)$$

where N is the total number of cells occupied by the total particles; ϕ_i is the local concentration of tracer particles in each cell; ϕ_m is the average concentration of tracer particles in the whole system. Theoretically, for a binary-mixture solid system, the maximum value of mixture variance, σ_0 , can be expressed by

$$\sigma_0^2 = \phi_m(1 - \phi_m). \quad (2)$$

The possible minimum variance in a cell, σ_r , is defined by

$$\sigma_r^2 = \frac{\phi_m(1 - \phi_m)}{n}, \quad (3)$$

where n denotes the possible number of particles in a cell. Then the Lacey mixing index M can be defined by

$$M = \frac{\sigma_0^2 - \sigma^2}{\sigma_0^2 - \sigma_r^2}. \quad (4)$$

Given the concentrations of the reference component in each cell, the mixing index can be defined and calculated by the statistical method. Lacey pointed out that this index is related to the number of the cells. Although it is not easy to represent the real mixing condition by one single index, the Lacey mixing index is still a good parameter for comparing the mixing status at different times or for different apparatus (Lacey, 1943). However this mixing index may be greater than 1 in some cases which causes some problems. This issue will be discussed further below.

(2) Kramer Mixing Index

Kramer proposed this mixing index (Kramer H.A., 1968) based on the same statistical variances as Lacey as in Eqs. (1–3):

$$M = \frac{\sigma_0 - \sigma}{\sigma_0 - \sigma_r}. \quad (5)$$

(3) Lacey, Weidenbaum and Bonilla Mixing Index (L. W.B. Mixing Index) (Weidenbaum S.S. and Bonilla C.F., 1953)

This index was first proposed for a horizontal cylindrical mixer. The mixing index M can be defined by

$$M = \frac{\sigma_r}{\sigma}, \quad (6)$$

where σ and σ_r are the same as defined in Eqs. (1) and (3).

(4) Mixing index by distance between the volume centers of two species (Avg. distance mixing index)

Two differently colored particles (with the same volumes) are often used to perform mixing/segregation experiments/simulations in academic studies. The distance (width or height, depending on the mixing/segregation direction) between the volume centers of each species is used to denote the mixing status. For example, **Fig. 1(a)** shows pre-organized particles (black on the left and white on the right) in a container. They are in a completely segregated state. The distance between the two volume centers is denoted by Δ_{seg} . **Fig. 1(b)** shows a similar case but the particles are organized so that black is on the bottom and white above. The distance between the volume centers of the two species in the horizontal direction (or in the vertical direction) is denoted as Δx . Thus the mixing index can be defined as follows:

$$M = \frac{\Delta_{\text{seg}} - \Delta x}{\Delta_{\text{seg}} - \Delta_{\text{mix}}}, \quad (7)$$

where Δ_{mix} is the value of Δx in a fully-mixed state which is usually 0 as shown in **Fig. 2**. This index is really based on the idea of the geometric positions of the particles as determined by identifying the volume (weight) center of particles. Calculation for this method would appear to be easy, but failure may occur because Δx could be 0 even when the particles are segregated, for example, if all the black particles were located in the upper-right and lower-left quadrants.

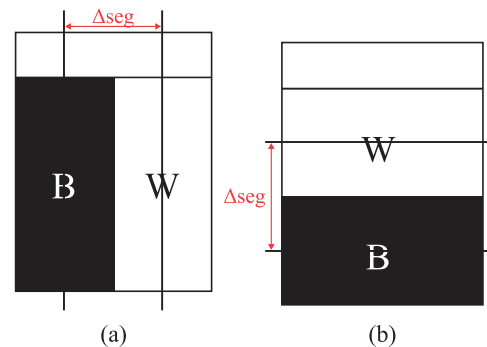


Fig. 1 Pre-organized particles with (a) black on the left and white on the right; (b) black on the bottom and white above in a container.

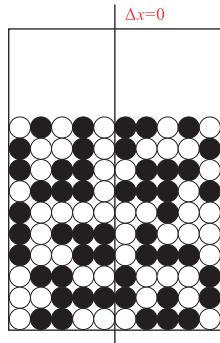


Fig. 2 Schematic representation of the fully-mixed particle state.

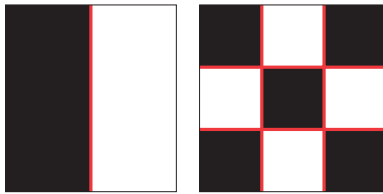


Fig. 3 The relation between the mixing degree and the length of the boundary between two different species.

3. Coordinate mixing index

As can be seen in **Fig. 3**, the mixing status (mixing degree or mixing index) has a strong relation with the length of the boundary between the two different species. The right configuration in **Fig. 3** has a better mixing condition than the left configuration, with the length of the boundary between the black-white areas of the right one being four times that of the left one. Based on the above idea, we try to develop a new mixing index, the coordinate mixing index, in this study.

An examination of **Fig. 4** is useful to explain how the coordinate mixing index is defined. Since the lengths of the contact interface between the black and white particles should be the same, whether counting either the white or the black particles, here we only use the white particles as the target. We can use the (coordinate) number of different-color contacts as the length, therefore the numbers inside each white circle (particle) denote the number of black particle(s) located in the neighboring position of the white particles. For example, in the lower left corner, the white particle has two black particles around it, so the coordinate number is 2. After counting the coordinate number for each white particle, the sum of all white particles is the total coordinate number for this system. The coordinate number for the granular system in **Fig. 4** is 17. A higher coordinate number indicates a better mixing status.

Let Ω denote the coordinate number of the system, and Ω_{ran} denote the coordinate number of the well-mixed (most random) system. Then the mixing index can be defined as

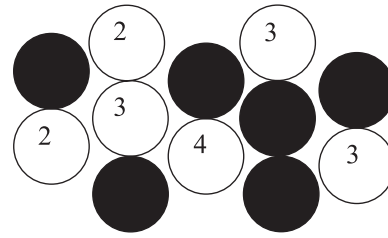


Fig. 4 Schematic representation of the method of calculation of the coordinate mixing index.

$$M = \frac{\Omega}{\Omega_{\text{ran}}} \quad (8)$$

If a cuboid box contains a , b , c particles in three directions (total abc particles), the coordinate number of the well-mixed arrangement would be

$$\Omega_{\text{ran}} = (a-1)bc + (b-1)ca + (c-1)ab \quad (9)$$

It is simpler to test and explain Eq. (9) using a 2D system with a and b particles distributed in the two directions. The right-hand image in **Fig. 3** represents a 3×3 particle system ($a = 3$, $b = 3$) in a well-mixed condition. For each row there are $a - 1 = 2$ coordinate numbers. With $b = 3$ rows, there would be $(a - 1) \times b = 6$ coordinate numbers. Similarly there are $(b - 1) \times a = 6$ coordinate numbers in the column direction. Therefore $\Omega_{\text{ran}} = (a - 1)b + (b - 1)a = 12$ coordinate numbers.

4. Results and discussions

In this study, we analyze the performance of different mixing indices with different examples of pre-organization, as in **Fig. 5**. Three different types of examples are examined: series A includes three different completely segregated initial system configurations; series B is devoted to investigating cases with different percentages of mixing particles; and in series C we investigate mixing starting with different special pre-organized patterns. The Lacey mixing index has been widely used in previous studies. However, the extent of the Lacey mixing index could be more than 1 in some cases. Therefore, we compare the Lacey mixing index and coordinate mixing index in a simulation of the granular mixing process in a vibrating bed.

Fig. 6 shows the degree of mixing obtained using different kinds of mixing indices for three different completely segregated initial configurations. In case I and case II, the degree of mixing with each mixing index is 0 except for the coordinate mixing index. This is because of the existence of a contact interface between the black and white particles in each case, which has to be counted when calculating the coordinate mixing index. In addition, although the Avg. distance mixing index is still indicates zero, the degree of mixing is calculated to be greater

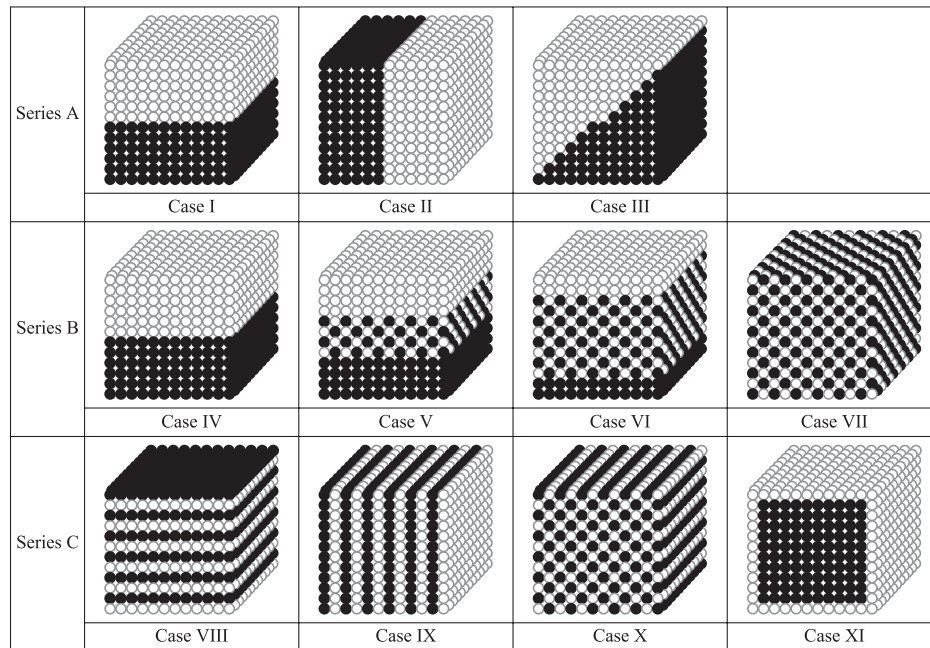


Fig. 5 Schematic representation of the configurations of black-white particles for different pre-arranged cases.

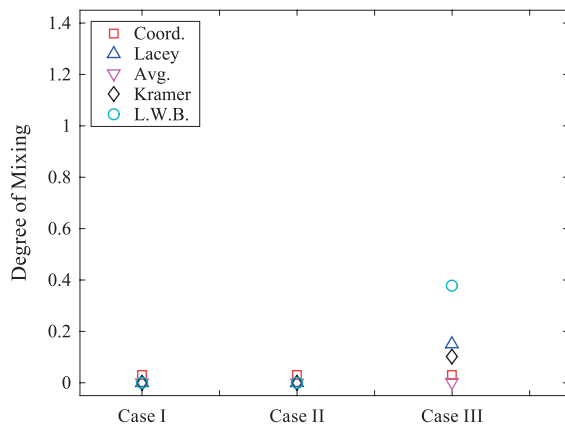


Fig. 6 The relation between the degree of mixing for different kind of mixing indices and different initial configurations of black-white particle systems. Total particle number of system is 1728; cell number N is 216; and number of particles (cell size) is $n = 8$.

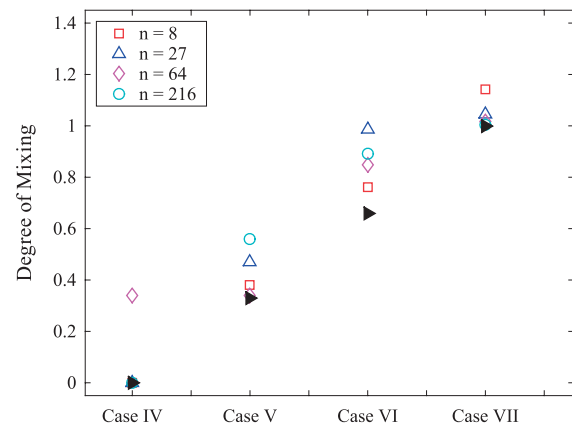


Fig. 7 The degree of mixing with the Lacey mixing index with different sizes of cells for series B. The cell number $N = 216$ for $n = 8$, $N = 64$ for $n = 27$, $N = 27$ for $n = 64$, and $N = 8$ for $n = 216$. Symbol fill: black, exact value.

than 0 in case III. Overall, the degree of mixing obtained with the L.W.B. mixing index is the highest while the Lacey Mixing Index is slightly larger than the Kramer Mixing Index.

Therefore, if a contact interface still exists between the black and white particles even in a well segregated situation, the degree of mixing obtained with the coordinate mixing index would not be zero. Differences in the length (number) of the interface also cause little deviation; the degree of mixing is similar. Meanwhile, the degree of mixing for the Avg. distance mixing index for series A is calculated to be zero. This is because the variances in the initial completely segregated state and the symmetry of the system both affect the calculated value.

The number and size of the cells should be chosen before using the Lacey mixing index to calculate the degree of mixing. If the size of the system is the same, the number of cells is dependent on the number of particles in a cell. For example, observation of series B shows 4 different configurations having 0 %, 33 %, 66 % and 100 % of particle mixing, respectively. The mixing states of the particle system are defined by four different volume ratios with fully mixed particles. For example, a mixing degree of 33 % means that 1/3 of the volume of particles is in a completely mixed condition in the system. **Fig. 7** shows the degree of mixing obtained with the Lacey mixing index for different sizes of cells for cases IV to VII. In case V, the calculated degree of mixing is close to the defined

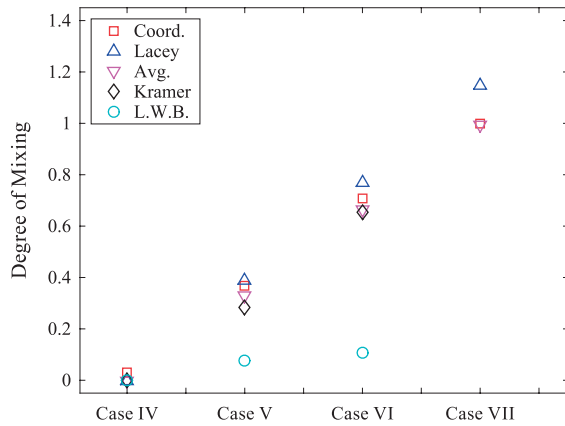


Fig. 8 The degree of mixing for different mixing indices for series B. The size of cell size is $n = 8$.

degree of mixing with $n = 64$. The error becomes larger while the cell size increases. However, this tendency does not occur in case VI. In case VII, the degree of mixing will be close to 1 when the cell size becomes larger. The degree of mixing even exceeds 1 with $n = 8$. In summary, accuracy is higher if we use a suitable cell size. The accuracy is highest for case V where all the white and black particles are well-mixed in each cell. This situation could not be achieved in case VI. Therefore, reducing the size of the cells could lead to variation in the system if we cannot find a suitable size of cell.

According to the results discussed above, $n = 8$ is chosen for all mixing indices, while the cell size varies in the formulation and the ensuing discussion. **Fig. 8** shows the degree of mixing obtained using different mixing indices for series B with a cell size of 8. We found the degree of mixing obtained with the Avg. distance mixing index to be closest to the pre-mixed value for this system. The error for the degree of mixing for the coordinate mixing index was almost 10 %, and for the Lacey mixing index it was 15 % in all cases. The degree of mixing obtained with the Kramer mixing index was either higher or lower than the pre-mixed value for different cases, but the value was abnormally high for case VII. We also found the degree of mixing to be larger than one for mixing indices based on the statistical method.

The degree of mixing obtained with different mixing indices for series C is shown in **Fig. 9**. In this figure, we can see that in addition to the coordinate mixing index, calculation with the other mixing indices shows a relatively high degree of mixing. From case VIII to case X, the degree of mixing calculated from the Lacey Mixing Index and Kramer Mixing Index is the same as for the well-mixed case, but this result is unreasonable. This is because if the number of different species of particle are the same in a cell, the mixing degree is indicated as well-mixed regardless of whether the different species of particle are mixed or not. The Avg. distance mixing index fails

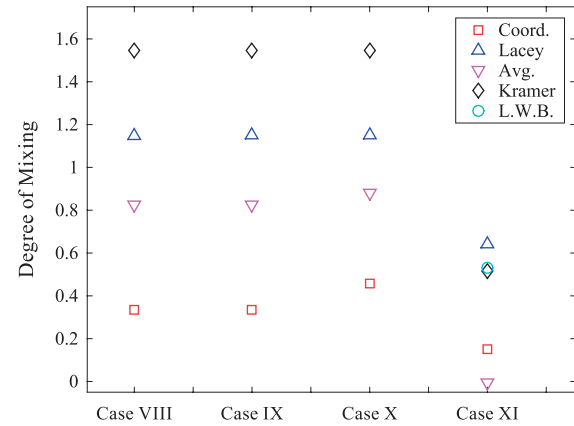


Fig. 9 The degree of mixing for different mixing indices for series C. The cell size is $n = 8$.

for case X and case XI, because the systems are symmetrical. In addition, since $\sigma = 0$ in each case for series C, the degree of mixing obtained with the L.W.B. mixing index is close to infinity. In these cases, this unreasonable situation does not happen when the coordinate mixing index is used. This also indicates that the coordinate mixing index has better accuracy and wider applicability.

From the above discussion, it can be seen that the degree of mixing obtained with the Lacey mixing index will be larger than one in some cases. This situation is contrary to expectations and is well known. The mixing behavior results in a situation between complete segregation and being well-mixed (the degree of mixing is between 0 and 1). Therefore, below we will discuss the mixing behavior of particles under different operating conditions. The dynamic behaviors of the particles were simulated by the discrete element method (DEM) in a vibrated granular bed, and the degrees of mixing were calculated with the Lacey mixing index and coordinate mixing index. The only difference between these two cases are the electrostatic force exist or not. The granular beds are energized by vertical sinusoidal oscillations at different vibrated acceleration and amplitude. The relevant operating conditions and parameters could be referred to the previous study (Lu L.S. and Hsiau S.S., 2005).

The evolution of the degree of mixing over time under different operating conditions is shown in **Figs. 10(a)** and **(b)**. We can see that both the trend and the final mixing degree for the different mixing indices are almost that same, as shown in **Fig. 10(a)**. In **Fig. 10(b)**, we consider the influence of the electrostatic force between the particles. To simplify the electrostatic effect, the two groups of glass beads are given opposite charges and the charge strength is assumed to be constant. In this case, the degree of mixing is greatly increased (Lu L.S. and Hsiau S.S., 2005). We found that the degree of mixing found with the Lacey mixing index would not exceed one, but this is unreasonable. At the same time, the final degree of

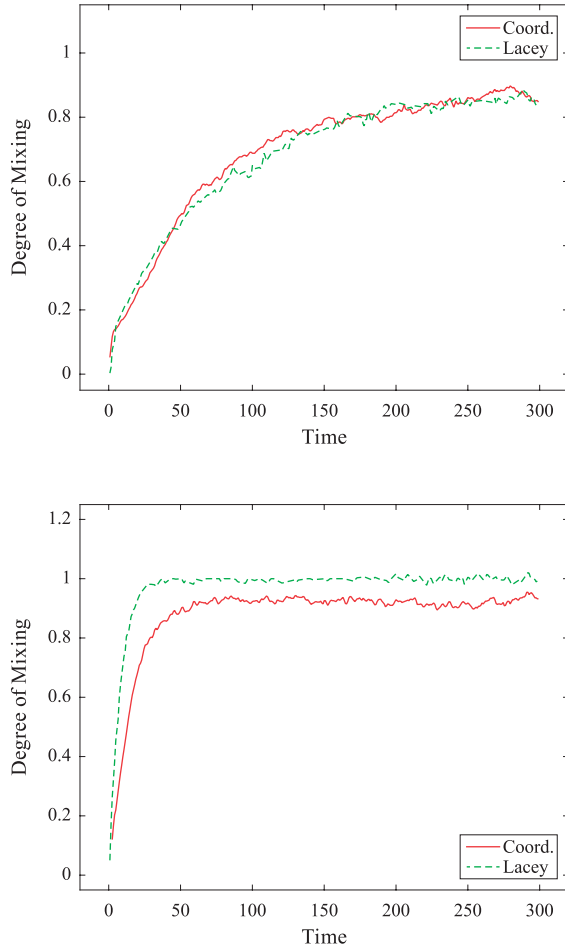


Fig. 10 Comparison of time evolution of mixing degree for the case where (a) $\Gamma = 3$, $f = 20$ Hz and the electrostatic force does not exist; and (b) $\Gamma = 5$, $f = 20$ Hz, but electrostatic force does exist. The dimensionless acceleration amplitude Γ was defined as $\Gamma = a/g$, where a is the acceleration; g was the acceleration due to gravity. The vibrational frequency is f .

mixing for the coordinate mixing index is quite stable. The final average mixing degree is about 0.95. If the overall concentration is about 0.5 in the system, the Lacey mixing index could be rewritten as

$$M = \frac{n}{n-1} \left(1 - \frac{\sigma^2}{0.25} \right), \quad (10)$$

and if the $M > 1$, the root mean square of the deviation σ is given by

$$\sigma^2 = \overline{(\phi_i - \phi_m)^2} < \frac{1}{4n}. \quad (11)$$

From the above equation, it can be seen that the size of the cell will affect the degree of mixing. Meanwhile, the mixing degree for the Lacey mixing index will be larger than one, if the variance σ^2 for the concentration of a reference component in each cell and overall concentration is smaller than a critical value.

5. Conclusions

In the granular mixing process, there are many factors that can affect the state of mixing in the system. The degree of mixing results also differ for different mixing indices. In this study, we discuss the performance obtained with different indices using different pre-organized particle configurations.

The results show that the cell size has a significant effect on the degree of mixing calculation given the criterion of the mixing index about a standard deviation. In some cases the degree of mixing for the Lacey mixing index, L.W.B. mixing index and Kramer mixing index will be larger than 1. Moreover, if the number of different species of particle in a cell is the same, the degree of mixing will be higher than the correct value regardless of how these different species of particle are mixed. In some cases, the Avg. distance mixing index will fail because the variation in the initial state and the symmetry of the system both affect this index.

In this study, we propose a new mixing index, called the coordinate mixing index. This index is relatively simple and widely applicable. According to the results obtained with different examples of pre-organized particles, we found that the degree of mixing obtained with the coordinate mixing index to always be between 0 and 1. Therefore, for now, the coordinate mixing index may be also a suitable index.

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Nomenclature

- f The vibrational frequency
- N Total number of cells occupied by the total particles
- n The possible number of particles in a cell
- Δx Distance between the volume centers of the two species in the horizontal direction (or in the vertical direction)
- Γ Dimensionless acceleration amplitude was defined as $\Gamma = a/g$, where a is the acceleration; g was the acceleration due to gravity
- Ω The coordinate number of the system
- ϕ_i Concentration of the reference component in the “ i th” cell
- ϕ_m Overall concentration of the reference component

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Author's short biography



Shih-Hao Chou

Dr. Shih-Hao Chou received his bachelor and PhD degrees in Department of Mechanical Engineering, National Central University, Taiwan in 2007 and 2013. He is currently a post-doctoral researcher at National Central University, Taiwan. His research interests include powder technology, mixing and segregation, DEM modeling of granular flow.



Yue-Lou Song

Mr. Yue-Lou Song received his bachelor in Department of Mechanical Engineering, National Central University, Taiwan in 2012. He served as an assistant of Department office (2008-2009). His research interests include mixing behavior in a vibrated granular bed.



Shu-San Hsiao

Shu-San Hsiao is an Associate Vice President for Research & Development of National Central University (NCU). He received his MS and PhD degrees from California Institute of Technology. He start his professorship life in NCU and majored in Mechanical Engineering in 1993. He also is a Distinguished Professor of Mechanical Engineering and Institute of Energy Engineering of NCU. His researches including Powder Technology, Clean Coal Technology, Modelling and Design of MOCVD, Energy Technology, Debris Flow and Avalanche, Hot-Gas Cleanup and Thermo-Fluids. He served as the chairman of Mechanical Engineering (2010–2013) and also as the director of Institute of Energy Engineering (2008–2013).